



福建師範大學
FUJIAN NORMAL UNIVERSITY

《机器学习》

【第六章-聚类】

距离度量

黄锐泓

本节要点

1. 聚类概述
2. 相似度与距离度量
3. 距离度量



聚类 (Clustering) 概述

1. 定义

聚类 (Clustering) 是**按照某个特定标准** (如距离) **把一个数据集分割成不同的类或簇**，使得同一个簇内的数据对象的相似性尽可能大，同时不在同一个簇中的数据对象的差异性也尽可能地大。也即聚类后同一类的数据尽可能聚集到一起，不同类数据尽量分离。



聚类 (Clustering) 概述

1. 定义

2. 聚类和分类

聚类：是指把相似的数据划分到一起，具体划分的时候并不关心这一类的标签，目标就是把相似的数据聚合到一起，聚类是一种 **无监督学习** (Unsupervised Learning) 方法。

分类：是把不同的数据划分开，其过程是通过训练数据集获得一个分类器，再通过分类器去预测未知数据，分类是一种 **监督学习** (Supervised Learning) 方法。



聚类 (Clustering) 概述

1. 定义
2. 聚类和分类
3. 聚类操作的核心——相似度的度量



相似度(Similarity)

1. 什么是相似度？

建模对象的相似/差异程度

相似度的度量是AI领域中重要的一环



相似度 (Similarity)

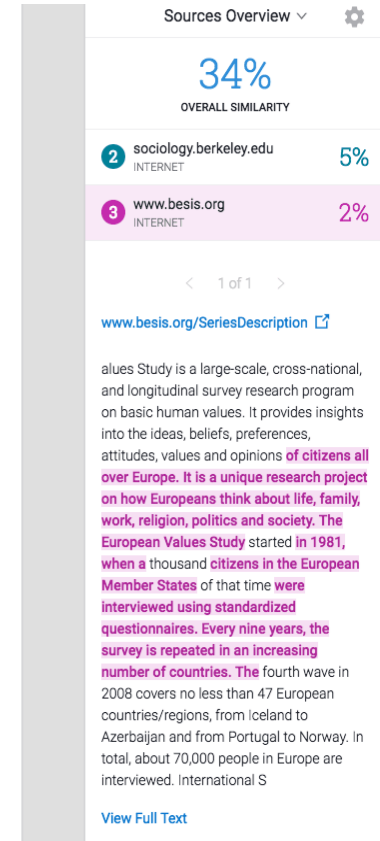
3. Data and Operationalization

3.1 Data

In order to test the hypotheses formulated in the previous chapter and eventually give a proper answer to the research question the data set that will be used is the European Value Study (2008), the European Values Study is a large-scale, time-intensive survey on basic human values. It provides insights into the values, beliefs and preferences of citizens all over Europe. It is a unique research project on how Europeans think about life, family, work, religion, politics and society. The European Values Study was launched in 1981, when a couple of hundred citizens in the European Member States were interviewed using standardized questionnaires. Every nine years, the survey is repeated in an increasing number of countries.

Not all the respondents of the original data sample are included in the analysis. People who did not answer one or more of the questions included, are filtered out of the dataset. The final number of respondent has been brought down to a sample analysis of 60077 respondents.

3.2 operationalization



相似度 (Similarity)

Camera 2



Camera 3



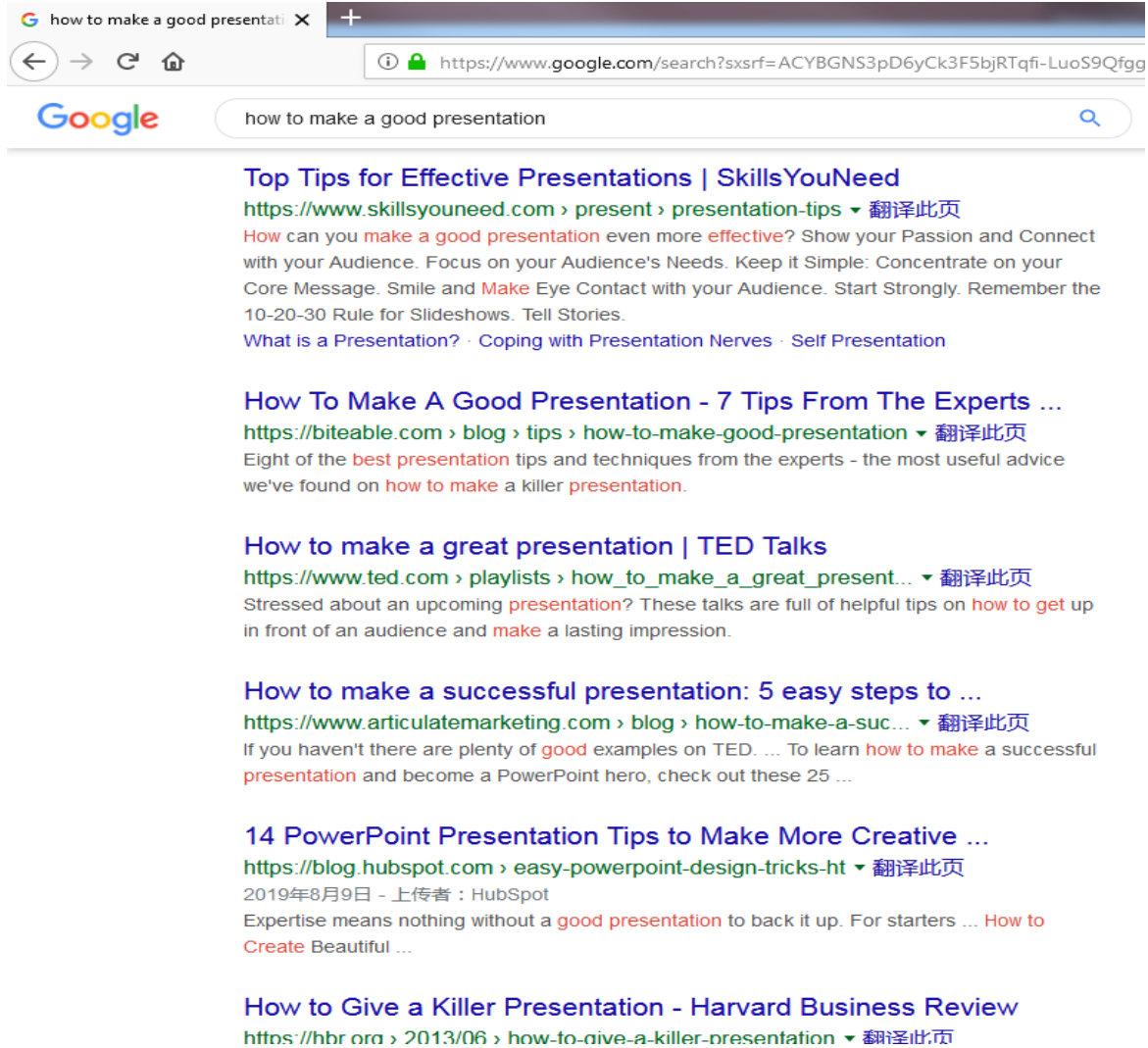
Camera 5



Camera 6



相似度 (Similarity)



The screenshot shows a Google search interface with the query "how to make a good presentation". The results list several articles with their titles, URLs, and brief descriptions. The first result is from SkillsYouNeed, the second from Biteable, the third from TED Talks, the fourth from Articulate Marketing, the fifth from HubSpot, and the sixth from Harvard Business Review.

how to make a good presentation X +

← → ↻ 🏠 <https://www.google.com/search?xsrf=ACYBGNS3pD6yCk3F5bjRTqfi-LuoS9Qfgg>

Google how to make a good presentation 🔍

Top Tips for Effective Presentations | SkillsYouNeed
<https://www.skillsyouneed.com/present/presentation-tips> ▼ 翻译此页
How can you **make a good presentation** even more **effective**? Show your Passion and Connect with your Audience. Focus on your Audience's Needs. Keep it Simple: Concentrate on your Core Message. Smile and **Make** Eye Contact with your Audience. Start Strongly. Remember the 10-20-30 Rule for Slideshows. Tell Stories.
What is a Presentation? · Coping with Presentation Nerves · Self Presentation

How To Make A Good Presentation - 7 Tips From The Experts ...
<https://biteable.com/blog/tips/how-to-make-good-presentation> ▼ 翻译此页
Eight of the **best presentation** tips and techniques from the experts - the most useful advice we've found on **how to make** a killer **presentation**.

How to make a great presentation | TED Talks
https://www.ted.com/playlists/how_to_make_a_great_present... ▼ 翻译此页
Stressed about an upcoming **presentation**? These talks are full of helpful tips on **how to get up** in front of an audience and **make** a lasting impression.

How to make a successful presentation: 5 easy steps to ...
<https://www.articulatemarketing.com/blog/how-to-make-a-suc...> ▼ 翻译此页
If you haven't there are plenty of **good** examples on TED. ... To learn **how to make** a successful **presentation** and become a PowerPoint hero, check out these 25 ...

14 PowerPoint Presentation Tips to Make More Creative ...
<https://blog.hubspot.com/easy-powerpoint-design-tricks-ht> ▼ 翻译此页
2019年8月9日 - 上传者: HubSpot
Expertise means nothing without a **good presentation** to back it up. For starters ... **How to Create** Beautiful ...

How to Give a Killer Presentation - Harvard Business Review
<https://hbr.org/2013/06/how-to-give-a-killer-presentation> ▼ 翻译此页



相似度 (Similarity)

1. 什么是相似度？
2. 如何计算物体间的相似度？

对物体进行建模，然后运用一点高中数学知识



相似度 (Similarity)

(1)



(4)



(2)



特征!!

(5)



(3)



相似度 (Similarity)

1. 什么是相似度？
2. 如何计算物体间的相似度？
3. 相似度与距离度量的关系

大多数情况下二者是相同的任务



距离度量(Distance Measurement)

通过距离比较两个对象的相似程度

回顾一下欧式距离

x, y两个对象的差距（距离）可以通过如下公式计算：

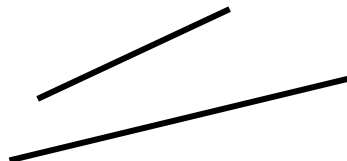
$$D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$



距离度量 (Distance Measurement)

欧式距离 (Euclidean Distance) $D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

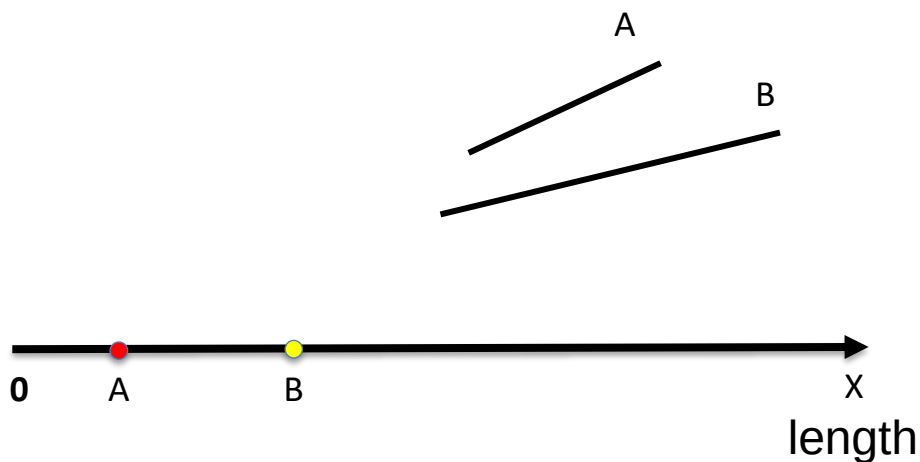
如何比较如下两个线段的相似度？



距离度量 (Distance Measurement)

欧式距离 (Euclidean Distance) $D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

如何比较如下两个线段的相似度？



$$\begin{aligned} D(x, y) &= \sqrt{(A_x - B_x)^2} \\ &= B_x - A_x \end{aligned}$$



距离度量(Distance Measurement)

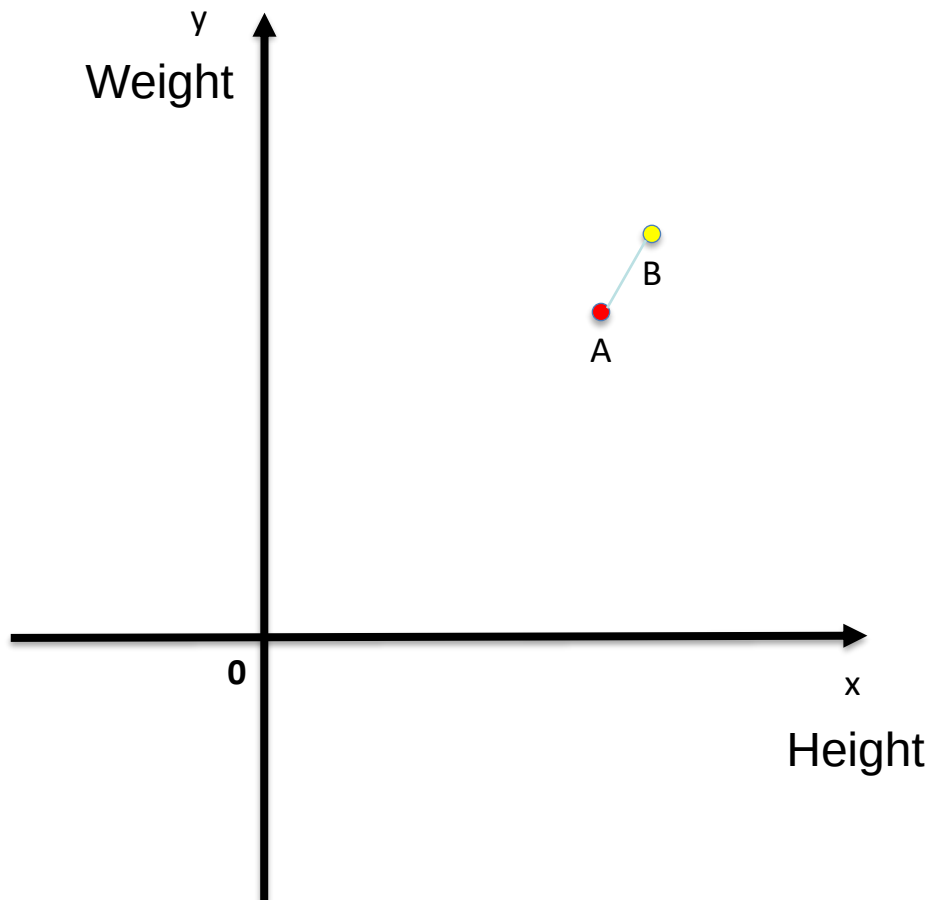
欧式距离(Euclidean Distance) $D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

如何比较两个人的相似度（如果只考虑身高体重的话）？



距离度量 (Distance Measurement)

欧式距离 (Euclidean Distance) $D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

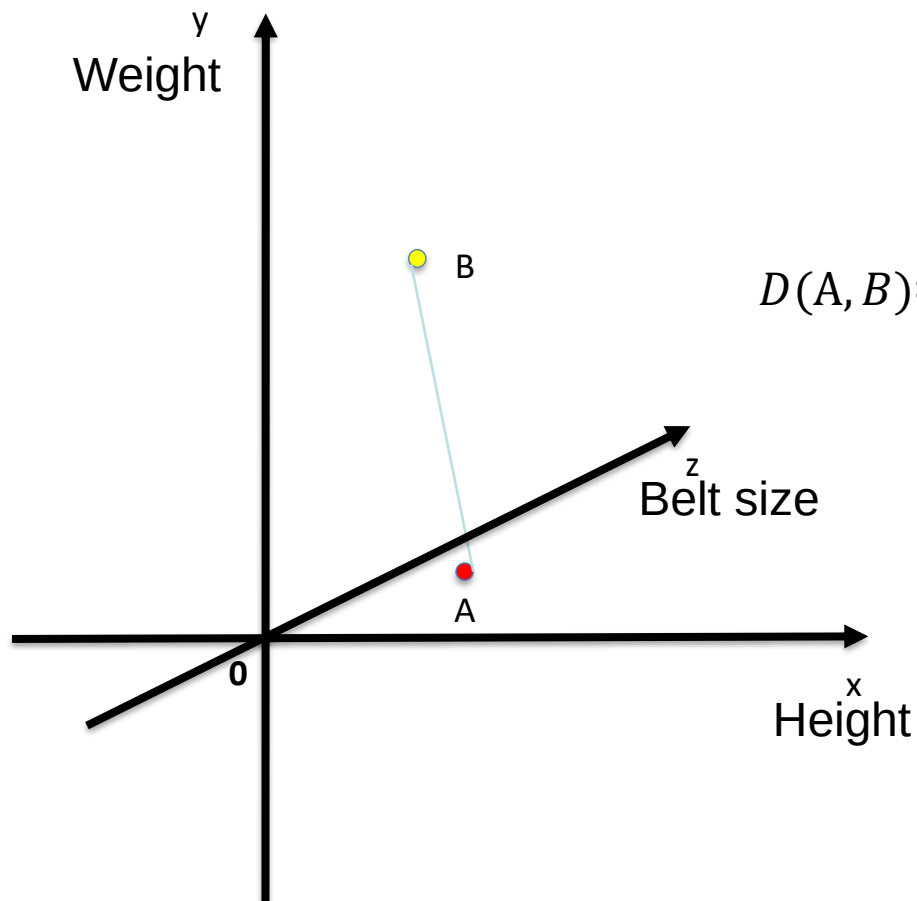


$$D(A, B) = \sqrt{(A_x - B_x)^2 + (A_y - B_y)^2}$$



距离度量 (Distance Measurement)

欧式距离 (Euclidean Distance) $D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$



$$D(A, B) = \sqrt{(A_x - B_x)^2 + (A_y - B_y)^2 + (A_z - B_z)^2}$$



距离度量(Distance Measurement)

闵可夫斯基(Minkowski Distance)

闵可夫斯基距离将样本看作高维空间中的点来进行距离的度量

设给定样本点的集合 D ，对于其中任意的 n 维向量 $X = (x_1, x_2, x_3, \dots, x_n)$ 和 $Y = (y_1, y_2, y_3, \dots, y_n)$ ，闵可夫斯基距离定义为：

$$D(X, Y) = (\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$$

其中 $p \geq 1$ 。



距离度量 (Distance Measurement)

闵可夫斯基 (Minkowski Distance) $D(X, Y) = (\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$

当 $p = 1$ 时，有 $D(X, Y) = \sum_{i=1}^n |x_i - y_i|$ ，此时又称为曼哈顿距离 (Manhattan Distance)，即绝对值之和。



距离度量(Distance Measurement)

闵可夫斯基(Minkowski Distance) $D(X,Y)=(\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$

曼哈顿距离 (Manhattan Distance) $D(X,Y) = \sum_{i=1}^n |x_i - y_i|$

曼哈顿距离通常称为出租车距离或城市街区距离，用来计算实值向量之间的距离。



距离度量(Distance Measurement)

闵可夫斯基(Minkowski Distance) $D(X,Y)=(\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$

曼哈顿距离 (Manhattan Distance) $D(X,Y) = \sum_{i=1}^n |x_i - y_i|$



距离度量 (Distance Measurement)

闵可夫斯基 (Minkowski Distance) $D(X, Y) = (\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$

当 $p = 2$ 时, 有 $D(X, Y) = (\sum_{i=1}^n |x_i - y_i|^2)^{\frac{1}{2}} = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$, 此时又称为欧氏距离 (Euclidean Distance), 即两点之间的直线距离。



距离度量(Distance Measurement)

闵可夫斯基(Minkowski Distance) $D(X,Y)=(\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$

当 $p = \infty$ 时，有 $D(X,Y)=\max_i |x_i - y_i|$ ，此时又称为切比雪夫距离

(Chebyshev Distance)，是各坐标数值差绝对值的最大值。



距离度量(Distance Measurement)

闵可夫斯基(Minkowski Distance) $D(X,Y)=(\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$

切比雪夫距离 (Chebyshev Distance) $D(X,Y)=\max_i |x_i - y_i|$

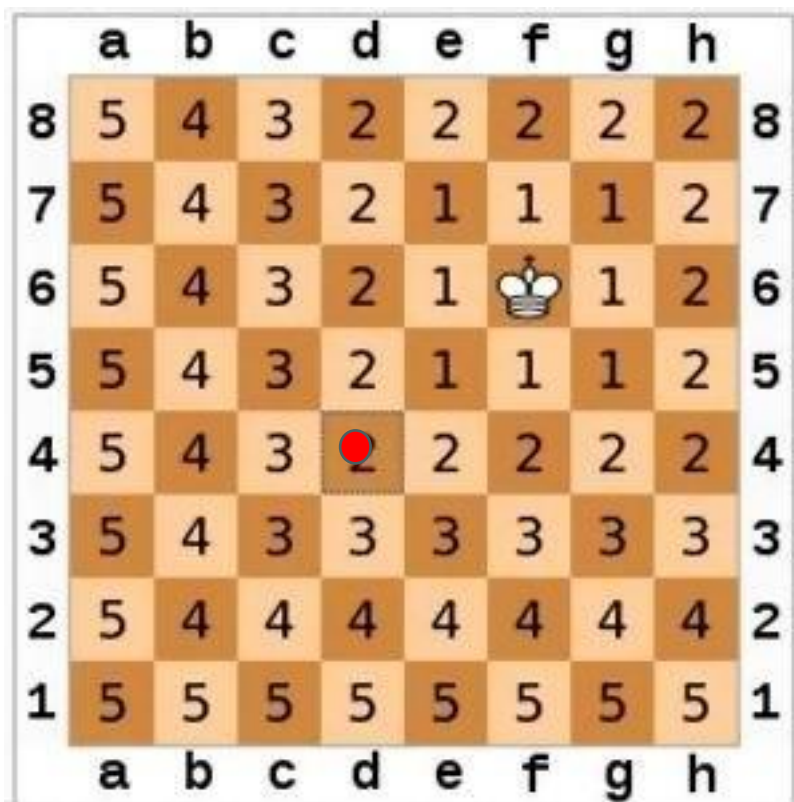
切比雪夫距离有时
又称为棋盘距离

	a	b	c	d	e	f	g	h	
8	5	4	3	2	2	2	2	2	8
7	5	4	3	2	1	1	1	2	7
6	5	4	3	2	1		1	2	6
5	5	4	3	2	1	1	1	2	5
4	5	4	3	2	2	2	2	2	4
3	5	4	3	3	3	3	3	3	3
2	5	4	4	4	4	4	4	4	2
1	5	5	5	5	5	5	5	5	1
	a	b	c	d	e	f	g	h	



距离度量(Distance Measurement)

闵可夫斯基(Minkowski Distance) $D(X,Y)=(\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$



将国王移到红点位置

切比雪夫距离为：2

曼哈顿距离为：4

欧式距离为： $2\sqrt{2}$



距离度量(Distance Measurement)

1. 两个点A和B的坐标分别为(7,3)、(4,9)如何计算他们的欧式距离、切比雪夫距离以及曼哈顿距离？
2. 两个点A和B的坐标分别为(1,3,6)、(2,4,5)如何计算他们的欧式距离、切比雪夫距离以及曼哈顿距离？

提示：闵可夫斯基(Minkowski Distance) $D(X,Y)=(\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$



距离度量(Distance Measurement)

两个点A和B的坐标分别为(1,4,6)、(2,3,5)如何计算他们的欧式距离、切比雪夫距离以及曼哈顿距离？

欧氏距离， $p = 2$:

$$D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

$$D = \sqrt{(1-2)^2 + (4-3)^2 + (6-5)^2}$$

$$D = \sqrt{1 + 1 + 1}$$

$$D = \sqrt{3}$$

切比雪夫距离， $p = \infty$:

$$D(X, Y) = \max_i |x_i - y_i|$$

$$D = \max\{|1-2|, |4-3|, |6-5|\}$$

$$D = \max\{1, 1, 1\}$$

$$D = 1$$

曼哈顿距离， $p = 1$:

$$D(x, y) = \sum_{i=1}^n |x_i - y_i|$$

$$D = \sum_{i=1}^3 |x_i - y_i|$$

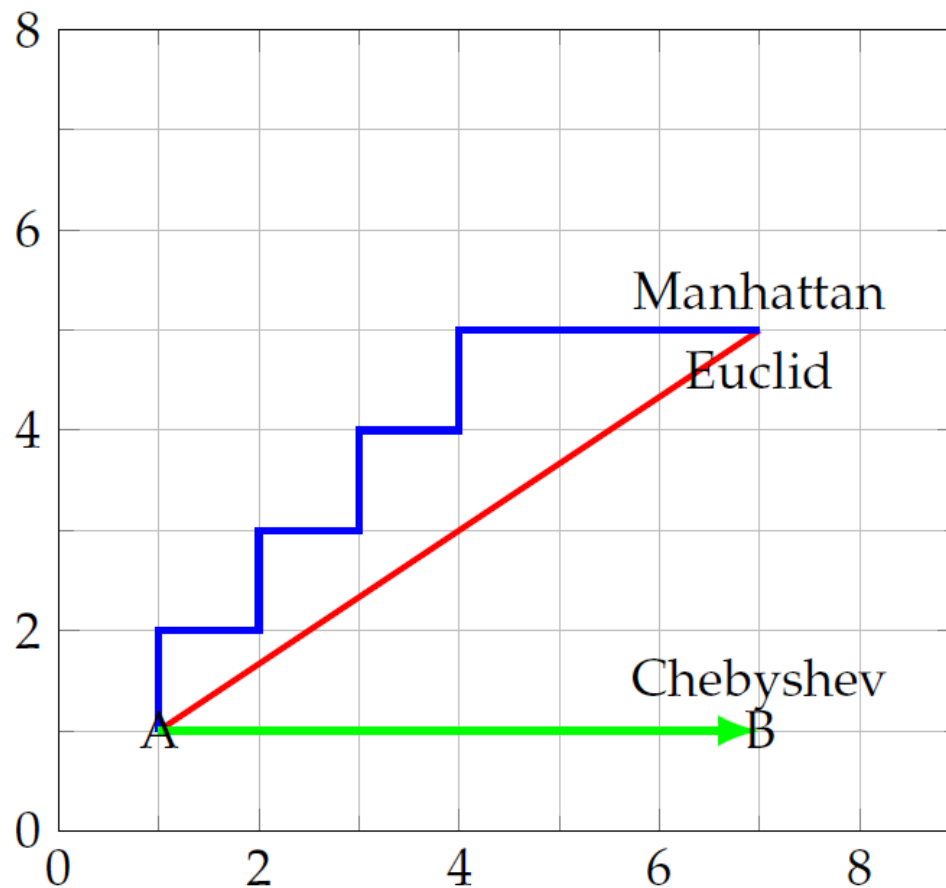
$$D = |1-2| + |4-3| + |6-5|$$

$$D = 3$$



距离度量 (Distance Measurement)

闵可夫斯基 (Minkowski Distance) $D(X, Y) = (\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$



距离度量(Distance Measurement)

余弦相似度(Cosine Similarity)

余弦相似度用向量空间中两个向量夹角的**余弦值**作为衡量两个个体间差异的大小。

相比距离度量，余弦相似度更加注重两个向量在**方向上的差异**，而非距离或长度上。

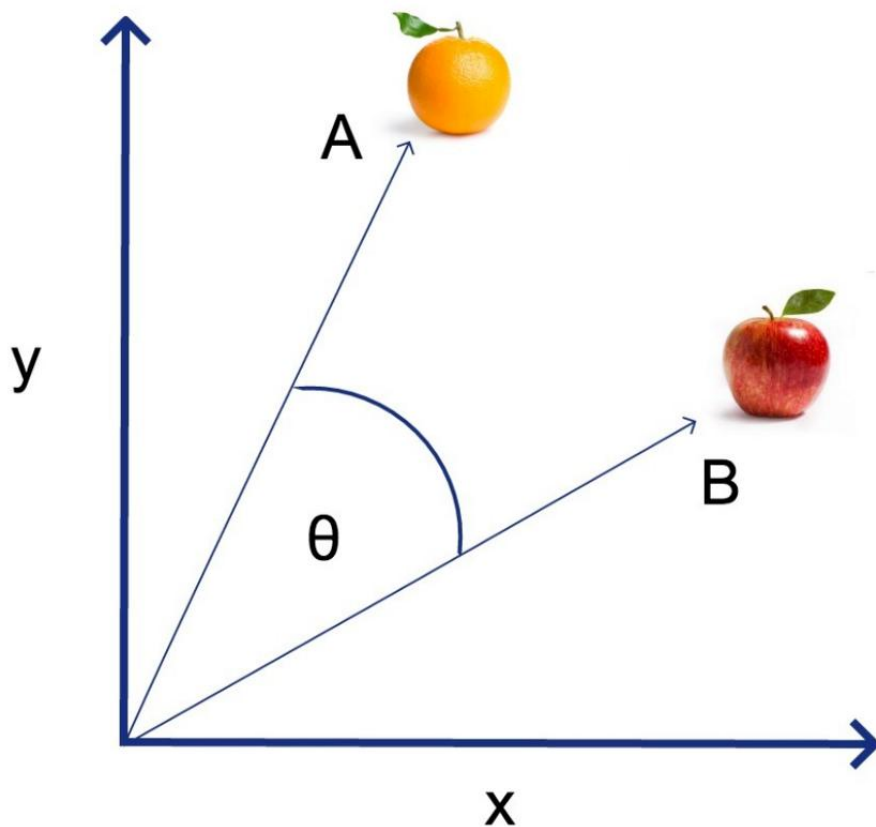
$$\text{sim}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n (A_i)^2 \sum_{i=1}^n (B_i)^2}}$$



距离度量 (Distance Measurement)

余弦相似度 (Cosine Similarity)

$$\text{sim}(A, B) = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n (A_i)^2 \sum_{i=1}^n (B_i)^2}}$$



距离度量(Distance Measurement)

马哈拉诺比斯距离(Mahalanobis Distance)

马氏距离，表示数据的**协方差**距离。生产环境中，变量之间往往存在一定的相关性，比如人的身高和体重，马氏距离能够同时考虑变量之间的**相关性**且又独立于尺度

A, B为两个
样本, Σ 为
协方差矩阵

$$D(A, B) = \sqrt{(\vec{A}_i - \vec{B}_i)^T \Sigma^{-1} (\vec{A}_i - \vec{B}_i)}$$



距离度量(Distance Measurement)

汉明距离(Hamming Distance)

汉明距离更多的用于信号处理，表明一个信号变成另一个信号需要的最小操作（替换位），实际中就是比较两个比特串有多少个位不一样。

$$D(A, B) = \sum_{i=1}^n I \{A_i \neq B_i\}$$



距离度量 (Distance Measurement)

汉明距离 (Hamming Distance)

$$D(A, B) = \sum_{i=1}^n I \{A_i \neq B_i\}$$

1011101 与 1001001 之间的汉明距离是 2

2143896 与 2233796 之间的汉明距离是 3

"toned" 与 "roses" 之间的汉明距离是 3



距离度量(Distance Measurement)

1. 分别计算1011101 与 1001011 , “karolin” 和 “kathrin” , 31738 和 32337 的汉明距离
2. 两个点A和B的坐标分别为(1,3,6)、(2,4,5)如何计算他们的余弦相似度?



距离度量(Distance Measurement)

1. 分别计算1011101 与 1001011 , “karolin” 和 “kathrin” , 31738 和 32337 的汉明距离

“karolin” 和 “kathrin” 两个字符串汉明距离为 3 (差异字符有: r-t,o-h,l-r),
1011101 和 1001011 汉明距离为 3(第3、5、6位不同),
31738 和 32337 汉明距离为 3 (不同整数位: 1-2,7-3,和 8-7).



距离度量(Distance Measurement)

2. 两个点A和B的坐标分别为(1,3,6)、(2,4,5)如何计算他们的余弦相似度？

余弦相似度：

$$D(x, y) = \frac{\sum_{i=1}^n (x_i y_i)}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}}$$

$$D = \frac{(2+12+30)}{\sqrt{1+9+36} \sqrt{4+16+25}}$$

$$D = \frac{44}{\sqrt{46} \sqrt{45}}$$



小结

1. 聚类概述
2. 相似度与距离度量
3. 距离度量(各种距离度量方法)



谢谢！

